Towards quantisation of Boussinesq solitons: Phase 2 master report (Routes A–E, Phases 2a–2h)

Project *quantum_boussinesq_solitons* — PI: Gustav Delius

Updated 4 November 2026

Abstract

Following the PI's initial decision, we first studied the *bad* Boussinesq equation

$$v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx},$$

whose soliton sector is regular and purely elastic, has regular breathers, and is the continuum limit of the quantum Toda chain (Phase 1), and subsequently extended the analysis to the *good* (well-posed) Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ and its Corrigan–Zambon integrable defects (Phases 2c–2h).

Semiclassical ($O(\hbar^{-1})$) results (Route A). Two-soliton scattering is an exactly symplectic map for the canonical structure of the field theory, *only* once ordering-dependent "topological" cross terms $d\Pi_i \wedge dM_j$ are included. These terms come from the kink in the potential ϕ. With them we obtain a closed-form semiclassical two-soliton phase, $S \simeq e^{i\delta/\hbar}$:

$$\delta(1,2) = \tfrac{2}{9}(\ell_1-\ell_2)\log A_{12} + \tfrac{2}{3}M_1M_2(v_1-v_2), \qquad \ell = \epsilon\cosh 3\beta \text{ (bad)}, \quad \ell = -\epsilon\cos 3\alpha \text{ (good)}.$$

- δ is antisymmetric (unitarity) and obeys a crossing-type relation under $\beta \to \beta + i\pi$.
- Continued to complex conjugate rapidities, $i\cdot\frac{2}{9}(\ell_1-\ell_2)$ reproduces *exactly* the adiabatic invariant of the Boussinesq breather computed independently by Bogdan and Charkina [1], turning breather quantisation into a string condition $\ell_1 - \ell_2 = -\frac{9i}{2}N\hbar$.

One-loop ($O(\hbar^0)$), Toda Fermi-edge, and defect-fusion synthesis (Routes A, D, E; Phases 2b–2h).

- *Candidate Gamma S-matrix tested and ruled out at $O(\hbar^0)$:* A natural Gamma-function ansatz $S \propto \frac{\Gamma(1\pm i\eta)}{\Gamma(1\mp i\eta)}$ in $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ (`smatrix_ansatz.pdf`, **(d)**) matches the semiclassical limit, unitarity, crossing, and the Bogdan–Charkina breather poles, but is *not exact*: it is ruled out at $O(\hbar^0)$ by the quantum Toda Fermi-edge limit (`toda_edge.pdf`), by the direct one-loop DHN calculation on the good equation (`oneloop_phase.pdf`), and by KP-triangle defect fusion (`oneloop_defect_phase.pdf`).
- *Exact one-loop Legendre χ_2 phase and canonical soliton momentum:* On the good equation, phonon scattering on N solitons and Corrigan–Zambon defects is reflectionless (`oneloop_phase.pdf`, `oneloop_defect_phase.pdf`). While δ at $O(\hbar^{-1})$ is elementary, the one-loop ($O(\hbar^0)$) phase evaluates in closed form to a 4-vertex KP coboundary of Legendre's dilogarithm $\chi_2(z) = \frac{1}{2}(\mathrm{Li}_2(z) - \mathrm{Li}_2(-z))$ plus a logarithmic coboundary. In the co-moving one-loop effective Lagrangian $L_1(v) = v\Delta P_{\mathrm{mode}} - \Delta E_1^{\mathrm{ren}}$, the light root $\phi_3 = \alpha$ drops out of $dL_1/d\alpha$ identically, yielding an $\alpha\ln\alpha$-free canonical collective-coordinate momentum $\Delta P_{\mathrm{can}}(\alpha,\epsilon) = -\frac{4\epsilon}{\pi}\alpha + O(\alpha^3)$ satisfying $d\Delta E_{\mathrm{can}} = v\, d\Delta P_{\mathrm{can}}$ (`charge_frame_master.pdf`).
- *Exact analytical match to the quantum Toda Fermi-edge limit:* In the quantum Toda Fermi-edge limit (`toda_edge_ba.py`, `matching.pdf`, `charge_frame_master.pdf`), measuring dressed momentum relative to the quantum Fermi edge absorbs the $\ln\hbar_T$ divergence into q, the classical frame term $+4\sqrt{u_1u_2} = -M_1\Pi_2/\hbar_T$ cancels the $-2/r$

Darboux singularity in $\partial\Phi_{\rm cl}/\partial u_2 = -2\ln\frac{1-r}{1+r}$, and the overtaking amplitude-ordering reversal ($v_{\rm good} = \cos\alpha$ vs $v_{\rm bad} = \cosh\beta$) flips $H_{\rm bad}(r) = -H_{\rm good}(r) = +\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$, leaving the exact $Y \to \infty$ residual $\lim_{Y\to\infty} R_0(Y, r) = -\frac{4}{\pi}(\chi_2(r) - r)$, which matches the numerical Phase 2f table to $< 7.1 \times 10^{-5}$.

- *Universal KP-Coboundary Master Function:* For any odd kernel $K(z) = -K(-z)$, the soliton–defect phase $\mathcal{I}_{SD}[K]$ and two-soliton phase $\mathcal{I}_{SS}[K]$ are the first and second KP-chord differences of a 2×2 dipole–dipole master function $\mathcal{F}_K(p, q)$ on the KP root space (`charge_frame_master.pdf`). We end with an updated ranked list of next steps towards the higher-loop / all-loop kernel $K(z; \hbar)$.

Contents

1 Setting **2**

2 Route A: semiclassical quantisation **3**
2.1 The symplectic structure of soliton scattering 3
2.2 The semiclassical phase 3
2.3 Bohr–Sommerfeld: breathers as spectral-parameter strings 4
2.4 What Route A gives and subsequent updates (Phases 2c, 2e, 2h) 4

3 Route B: quantum inverse scattering on the line **5**

4 Route C: W_3, sl_3 affine Toda and the $U_q(\widehat{\mathfrak{sl}}_3)$ R-matrix **6**

5 Route D: discrete models and the quantum Toda Fermi-edge limit **6**

6 Route E: integrable defects and the Universal KP-Coboundary Master Function **8**

7 Consistency checks across candidates and orders **9**

8 Conclusions and ranked open problems **9**

A Phase 2 Write-ups and Verification Scripts **10**

1 Setting

Throughout, $v = 2(\log\tau)_{xx}$ and $k = \sinh\beta$. A soliton has velocity $v = \epsilon\cosh\beta$, with $\epsilon = +1$ for right-movers and $\epsilon = -1$ for left-movers. The potential is $v = \phi_x$ and the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \tfrac{1}{2}\phi_{xx}^2 + \phi_x^3,$$

with $\pi = \phi_t$ and canonical symplectic form $\Omega = \int \delta\pi \wedge \delta\phi$.

On one soliton the charges are:

- mass $M = \int v = 2k$;
- ϕ-shift charge $\Pi = \int \pi = -vM$, conserved because $\phi \to \phi + c$ is a symmetry;
- momentum $P = -\int \pi\phi_x = \frac{2}{3}\epsilon k^3\cosh\beta$;
- energy $E = \frac{2}{15}k^3(5 + 4k^2)$.

These satisfy $dE/dP = v$. **(a)** [1]/**(c)**

The two-soliton constants are

$$A_{\mathrm{ov}} = \frac{1 + 3\coth^2 \frac{\beta_1+\beta_2}{2}}{1 + 3\coth^2 \frac{\beta_1-\beta_2}{2}}, \qquad A_{\mathrm{ho}} = \frac{1 + 3\tanh^2 \frac{\beta_1-\beta_2}{2}}{1 + 3\tanh^2 \frac{\beta_1+\beta_2}{2}},$$

both in $(0, 1)$, and the position shifts are $\Delta x_1 = -k_1^{-1} \log A$, $\Delta x_2 = +k_2^{-1} \log A$ (soliton 1 starts on the left). **(c)** (Phase 1)

What a soliton S-matrix would mean here. The model has no boost symmetry and is gapless (phonons with $\omega \simeq |k|$). The natural object is therefore the non-relativistic two-body S-matrix $S(P_1, P_2)$ of soliton asymptotic states, in the sense of Faddeev–Korepin [7]. Classical scattering is purely transmissive and diagonal, so S is a phase; for two species (left and right movers) S is diagonal.

For diagonal scattering the Yang–Baxter equation is automatic. The non-trivial requirements are:

- unitarity;
- the correct semiclassical limit;
- analyticity and a crossing substitute;
- a bound-state (breather) spectrum consistent with poles.

2 Route A: semiclassical quantisation

2.1 The symplectic structure of soliton scattering

Proposition 2.1 (**(c)**). *Consider two well-separated solitons, with ϕ normalised by $\phi(-\infty) = c$. Up to exponentially small terms,*

$$\Omega = d\Pi_{\mathrm{tot}} \wedge dc + \sum_j dP_j \wedge dX_j + \big[X_i > X_j\big]\, d\Pi_i \wedge dM_j.$$

The argument is as follows. π_i is localised at X_i, and there ϕ_j equals 0 or M_j according to whether soliton i is to the left or right of soliton j. The self-term $dP \wedge dX$ was derived in Phase 1. In sine-Gordon the analogous cross term vanishes, because the kink height $2\pi/\beta$ is fixed. Here $M = 2k$ varies with the soliton parameter.

Proposition 2.2 (**(c)**, `scripts/symplectic_scattering.py`). *The classical scattering map $X_j \mapsto X_j + \Delta x_j$ (with k_j and c fixed) preserves Ω, i.e.*

$$\sum_j dP_j \wedge d\Delta x_j = (\textit{cross term})_{\mathrm{in}} - (\textit{cross term})_{\mathrm{out}}.$$

This was verified to 40 digits for overtaking and head-on collisions at five parameter points each. It was also verified for the good *equation in its regular region (`scripts/consistency_checks.py`). Without the cross terms the identity fails by $O(1)$ amounts; for example, the left side equals* 19.79 *at $\beta = (0.9, 0.3)$.*

This is a non-trivial confirmation of the classical picture. It also shows that a semiclassical S-matrix built from the naive rule "$\Delta x_j = -\partial\delta/\partial P_j$" does not exist: the shifts are not curl-free in the canonical momenta.

2.2 The semiclassical phase

In each ordering region we choose Darboux coordinates by the rule

$$X'_j = X_j + \gamma_j \sum_{i \text{ left of } j} M_i, \qquad \gamma_j = \frac{d\Pi_j/dk_j}{dP_j/dk_j}.$$

The mirrored rule (normalising ϕ at $+\infty$) gives the same in–out differences. The shifts $\Delta x'_j$ are then curl-free. Their potential is the semiclassical phase δ, with $\Delta x'_j = -\partial\delta/\partial P_j$ and $S \simeq \exp(i\delta/\hbar)$, by stationary phase on two-particle wave packets [7].

Proposition 2.3 (**(c)**, `scripts/semiclassical_phase.py`, `scripts/consistency_checks.py`). *1. $\frac{dP}{d\beta} = \frac{2}{3}\,k\sinh 3\beta$ for the bad equation. For the good equation, $\frac{dP}{d\alpha} = \frac{2}{3}\,k\sin 3\alpha$ with $k = \sin\alpha$; this vanishes at the cusp $\alpha = \pi/3$ found in Phase 1.*

2. For both overtaking and head-on collisions, and both equations,

$$\boxed{\delta(1,2) = \tfrac{2}{9}\,(\ell_1 - \ell_2)\,\log A_{12} + \tfrac{2}{3}\,M_1 M_2\,(v_1 - v_2)} \tag{1}$$

with $\ell = \epsilon\cosh 3\beta$ (bad) or $\ell = -\epsilon\cos 3\alpha$ (good). The remainder was first found by symbolic integration, then identified as $\frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$ by exact matching.

3. Unitarity: $\delta(1,2) = -\delta(2,1)$.

4. Crossing-type relation: $\delta_{\text{ov}}(\beta_1, \beta_2 + i\pi) = -\delta_{\text{ho}}(\beta_1, \beta_2)$. *The shift $\beta \to \beta + i\pi$ maps a right-mover to a left-mover with $M \to -M$.*

Remark 2.4.

- ℓ is proportional to the spectral parameter of the third-order Lax operator. In the uniformisation of [4], $\lambda = \frac{1}{2}(k^3 + k^{-3})$, with k^3 playing the role of $e^{3\beta}$.
- The dynamical part of δ is (difference of spectral parameters)$\times \log A$.
- The second part is a "mutual statistics" phase: an Aharonov–Bohm-like phase $\propto \Pi_1 M_2 - \Pi_2 M_1$ between objects carrying ϕ-shift charge and ϕ-winding. **(d)** (interpretation)

Remark 2.5 (KdV limit, **(c)**). For small amplitudes the topological term is of the *same order* as the KdV-type term $2\beta_1 \log A$ in $\partial\delta/\partial\beta_1$. Numerically, their ratio tends to 0.564 for $\beta_1/\beta_2 = 1.7$. The heuristic of Bonnemain and Doyon [2] identifies the quantum phase derivative with $P'_{\text{GHD}}\Delta$, with $P_{\text{GHD}} = \eta^2/2$. That identification reproduces our dynamical term but *misses the topological term.* The classical generalised hydrodynamics, which uses only Δx, is unaffected. **(c)**

2.3 Bohr–Sommerfeld: breathers as spectral-parameter strings

Bogdan and Charkina quantise the Tajiri–Murakami breather of the bad equation by Bohr–Sommerfeld, $I_{\text{br}} = N\hbar$, with an independently computed adiabatic invariant I_{br} **(a)** [1, (64)–(65)].

Proposition 2.6 (**(c)**). *Set $\beta_{1,2} = a \pm ib$, which is Bogdan–Charkina's parametrisation $\kappa \pm ik = \sinh(a \pm ib)$. Then:*

1. *A_{12} equals their a_{12} [1, (33)]. Regularity $A > 1$ holds if and only if $b > \pi/3$, which is their existence boundary.*
2. *$i \cdot \frac{2}{9}(\ell_1 - \ell_2) = I_{\text{br}}$ exactly, with $I_{\text{br}} = -\frac{4}{9}\sinh 3a \sin 3b$, in agreement with [1, (64)].*

Hence the coefficient of $\log A$ *in* (1) *is the breather action. Bohr–Sommerfeld quantisation reads*

$$\ell_1 - \ell_2 = -\tfrac{9i}{2}\, N\hbar, \qquad N \in \mathbb{N}, \tag{2}$$

which is a string condition in the spectral parameter ℓ*.*

This is the strongest cross-check in this report. It ties our two-soliton phase, built from Hirota shifts and the canonical symplectic form, to an independent quasiclassical calculation in the literature. Item 2 was checked at three parameter points to 10^{-16}.

Equation (2) is the form in which bound states appear in Bethe-ansatz solvable models. It suggests that ℓ is the variable in which an exact S-matrix has poles: $\ell_1 - \ell_2 \in -\frac{9i}{2}\hbar\mathbb{N}$, possibly with $\hbar$ renormalised. **(d)**

2.4 What Route A gives and subsequent updates (Phases 2c, 2e, 2h)

- *Obtained at tree level (*$O(\hbar^{-1})$*).* The leading semiclassical phase (1), which is unitary, crossing-type covariant and consistent with breather quantisation. Also an asymptotic Bethe-ansatz quantisation of N solitons on a ring, $P_j\Lambda + \sum_{k\neq j}\delta(j,k) = 2\pi\hbar n_j$. **(d)**: the twisted boundary conditions forced by ϕ-winding on a ring have not been worked out. Unlike in the relativistic Toda case [12, 13], the tree-level phase δ is *elementary*: no dilogarithm appears at $O(\hbar^{-1})$, because $dP \propto k\sinh 3\beta\, d\beta$ and integration by parts closes.

- *Update (Phase 2c,* `smatrix_ansatz.pdf`*): Candidate Gamma ansatz.* Setting $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ and $R(x) = \Gamma(1+ix)/\Gamma(1-ix)$, a candidate exact S-matrix was proposed **(d)**: $S = R(\eta)(e^2A/\eta^2)^{i\eta}e^{iT/\hbar}$ (overtaking) and $S = R(\eta)^{-1}(A\eta^2/e^2)^{i\eta}e^{iT/\hbar}$ (head-on). Its semiclassical expansion gives $\arg S - \delta/\hbar \to \pm\frac{\pi}{2}\operatorname{sgn}\eta + O(\hbar)$ and its poles match the Bogdan–Charkina breather levels (2).

- *Update (Phase 2d,* `oneloop.pdf`*): Bad-equation one-loop DHN analysis.* For the bad equation, modes with $|q| > 1$ are exponentially unstable ($|A(\mathrm{sol},\mathrm{ph})| \neq 1$), so the DHN sum is well posed only on the Charlier–Lenells invariant stable subspace $|q| < 1$ (Scheme A). There the breather one-loop energy and momentum are holomorphic continuations of the constituent soliton values, so the string condition (2) receives no $O(\hbar^0)$ shift at fixed rapidity; however, a reversed-group-velocity mode at $q^* = -\sqrt{3}/2$ is captured by slow solitons, making Scheme A differ from the Toda continuum limit at $\beta \to 0$. **(c)**

- *Update (Phase 2e,* `oneloop_phase.pdf`*): Good-equation one-loop two-soliton phase.* For the well-posed good equation ($k = \sin\alpha$, $v = \epsilon\cos\alpha$), the 4×4 linearised operator is reflectionless for all $q \in \mathbb{R}$, with a 3-root continuum phase shift $\theta_S'(q;\alpha,\epsilon) = \sum_{r=1}^{3} c_r\Theta'(q;\phi_r^{(\epsilon)})$, weights $(c_1,c_2,c_3) = (1,1,-2)$, and angles $(\phi_1,\phi_2,\phi_3) = (\alpha+2\pi/3, \alpha-2\pi/3, \alpha)$. By Levinson's theorem, the mode count jumps at the cusp $\alpha = \pi/3$ where a heavy-branch instability appears. On the two-soliton background, the Faddeev–Korepin one-loop phase evaluates in closed form to a 3×3 sum over $L(z) = -2\chi_2(z) - \ln|z|\ln\left|\frac{1-z}{1+z}\right|$, where $\chi_2(z) = \frac{1}{2}(\mathrm{Li}_2(z) - \mathrm{Li}_2(-z))$ is Legendre's dilogarithm! Thus **dilogarithms enter at one loop ($O(\hbar^0)$) rather than at tree level ($O(\hbar^{-1})$)**, ruling out both the continued Gamma candidate (A) and the pure exponential (B) as exact. **(c)**

- *Update (Phase 2h,* `charge_frame_master.pdf`*): One-loop soliton charge audit.* Because the Boussinesq soliton profile $u_0(x-vt;\alpha)$ changes width $k = \sin\alpha = \sqrt{1-v^2}$ with velocity $v = \epsilon\cos\alpha$, the fixed-profile mode-sum charges $(\Delta E_1^{\mathrm{ren}}, \Delta P_{\mathrm{mode}})$ have a non-zero velocity anomaly $\mathcal{A}_v(\alpha) = d\Delta E_1^{\mathrm{ren}}/d\alpha - v\, d\Delta P_{\mathrm{mode}}/d\alpha \neq 0$. In the co-moving one-loop effective Lagrangian $L_1(v) = v\Delta P_{\mathrm{mode}} - \Delta E_1^{\mathrm{ren}}$, the total weight $\cos(2\phi_r) - \cos(\alpha+\phi_r)$ vanishes

identically on the light root $\phi_3 = \alpha$, cancelling all $\ln\alpha$ singularities to all orders and giving the canonical collective-coordinate momentum **(c)**

$$\Delta P_{\rm can}(\alpha,\epsilon) = \frac{\partial L_1}{\partial v} = -\epsilon\frac{\sqrt{3}\cos\alpha}{\pi}\ln\left|\frac{\tan\frac{\alpha+2\pi/3}{2}}{\tan\frac{\alpha-2\pi/3}{2}}\right| = -\frac{4\epsilon}{\pi}\alpha + \frac{8\epsilon}{9\pi}\alpha^3 - \frac{8\epsilon}{45\pi}\alpha^5 + O(\alpha^7),$$

which satisfies $d\Delta E_{\rm can}/dv = v\, d\Delta P_{\rm can}/dv$ identically.

3 Route B: quantum inverse scattering on the line

What would have to be true. The Lax matrix $U(x,\lambda)$ would need an ultralocal r-matrix Poisson algebra, so that it can be regularised on a lattice and quantised à la Faddeev–Korepin–Sklyanin. Then a Bethe ansatz over a pseudo-vacuum with soliton excitations would give S.

Proposition 3.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *In canonical variables the Lax data of Phase 1 are* $q = \frac{3}{2}\phi_x - \frac{1}{4}$ *and* $r = \frac{1}{2}q_x + \frac{\sqrt{3}}{4}\pi$ *(good-equation real form; the bad one differs by* i*'s). Their brackets are*

$$\{q(x), r(y)\} = \tfrac{3\sqrt{3}}{8}\,\delta'(x-y), \qquad \{q,q\} = \{r,r\} = 0.$$

So the companion-matrix Lax connection is non-ultralocal*.*

Where it stalls. There are two known ways out, and neither yet gives a theory on the line.

1. Maillet's r/s formalism with lattice regularisation [14]. No construction exists for Boussinesq.
2. The Miura (free-field) map. This is the route of Bazhanov–Hibberd–Khoroshkin [3] and Hikami [10]. It yields $U_q(\widehat{\mathfrak{sl}}_3)$ T- and Q-operators, but in the chiral conformal setting on a circle, with no particle sector.

Dimakis et al.'s Yang–Baxter maps [6] concern polarisations in the matrix case and are vacuous for the scalar equation. The "beyond Yang–Baxter" remark there is therefore not an obstruction here. For diagonal scattering, Yang–Baxter holds trivially. **(c)**

4 Route C: W_3, sl_3 affine Toda and the $U_q(\widehat{\mathfrak{sl}}_3)$ R-matrix

What would have to be true. Boussinesq solitons would have to be identifiable as a limit of $a_2^{(1)}$ (complex) affine Toda solitons, whose S-matrices are the $U_q(\widehat{\mathfrak{sl}}_3)$ intertwiners of Hollowood [12] and Gandenberger [8].

Proposition 4.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *Take Hollowood's* $a_2^{(1)}$ *interaction constant [11, (1.19)],* $A^{(12)}(\theta) = \frac{2\cosh\theta+1}{2(1+\cosh\theta)}$*, between the two fundamental species. With* $S = \beta_1+\beta_2$ *and* $D = \beta_1 - \beta_2$*,*

$$A^{\rm bad}_{\rm ov}(\beta_1,\beta_2) = \frac{A^{(12)}(S)}{A^{(12)}(D)}\cdot\frac{\tanh^2(D/2)}{\tanh^2(S/2)}, \qquad A^{\rm bad}_{\rm ho}(\beta_1,\beta_2) = \frac{1}{A^{\rm bad}_{\rm ov}(\beta_1,\beta_2+i\pi)}.$$

Here $\tanh^2(\theta/2)$ *is the sine-Gordon soliton–soliton constant. Equivalently,*

$$A_{\rm ov} = \frac{G(S/2)}{G(D/2)}, \qquad G(y) = \frac{16\prod_{a=1,2}\sinh\frac{y+ia\pi/3}{2}\sinh\frac{y-ia\pi/3}{2}}{\sinh^2 y}.$$

So the Boussinesq interaction is built from exactly the $\mathfrak{sl}_3$ blocks, but in the form "$F(\beta_1 - \beta_2)/F(\beta_1 + \beta_2)$". This is the structure of scattering off a particle *and its mirror image*. It suggests that a Boussinesq S-matrix might take the form

$$S(\beta_1, \beta_2) = \mathcal{S}(\beta_1 - \beta_2)\, \mathcal{S}(\beta_1 + \beta_2 + i\pi)^{\pm 1}$$

with $\mathcal{S}$ built from $U_q(\widehat{\mathfrak{sl}}_3)$ and sine-Gordon factors. Such a form is reminiscent of boundary (reflection) amplitudes. That would connect to the "boundary affine Toda" structure in Bazhanov–Hibberd–Khoroshkin [3]. **(d)**

The obstruction is that the semiclassical phase (1) is not of this product form. Its weight $\frac{2}{9}(\ell_1 - \ell_2)$ is not a sum of functions of S and D, and its topological part has no Toda analogue.

Further obstructions:

- Toda solitons are relativistic with fixed masses m_a. In the Toda constant the mass term $4m^2 \sin^2(\pi a/3)$ appears where Boussinesq has K^4, so the species label is "replaced" by the wavenumber. There is no limit of fixed-species Toda solitons that yields Boussinesq solitons. **(c)**
- Takács and Watts show that $a_2^{(1)}$ soliton–breather scattering is flawed both classically and quantum mechanically **(b)** [20].

Route C stalls: the structural similarity is precise, but no dictionary is available. What is lost in the non-relativistic reduction is the difference form, and with it the standard crossing.

5 Route D: discrete models and the quantum Toda Fermi-edge limit

(i) Quantum discrete Boussinesq (Hikami [10]). This is the Belov–Chaltikian lattice W_3 chain, periodic, with a Baxter Q-operator. A soliton/particle sector would require a thermodynamic limit with an identified ground state and excitations. That has not been done in the literature, and we did not attempt it. *Stalls immediately*; it is a research project in itself.

(ii) Quantum Toda chain. This is our recommended route for the small-amplitude corner. The required ingredients are:

1. The exact quantum Toda chain **(a)** [9, 17] and its asymptotic Bethe ansatz **(a)/(b)** [19, 15, 18]. Its particle excitations become solitons in the classical limit **(b)** [15, 16].
2. The dressed particle–particle S-matrix in the finite-density ground state.
3. Its classical limit should reproduce the Toda soliton phase.
4. The small-amplitude scaling limit should give (1).

Update (Phase 2b, `toda_route.pdf`). Steps 2–4 were carried out in the classical limit, and step 2 at finite $\hbar$ to $O(\hbar)$. **(c)**

- The classical limit of the Bethe ansatz reproduces the Toda soliton energies, momenta and exact two-soliton shifts.
- Its two-body phases carry exactly the predicted topological cross terms.
- The label-frame phase is $\delta_T = (p_1 - p_2)\log A_T - (\Pi_1 M_2 - \Pi_2 M_1)$.
- In the scaling limit, $\delta_T \to 3b\,\delta_{\text{bad}}$ for overtaking *and* head-on collisions. This independently confirms (1), including its topological term, from an exactly solvable quantum model.

Update (Phases 2d, 2f, 2h, `toda_edge.pdf`, `matching.pdf`, `charge_frame_master.pdf`): Fermi-edge double-scaling limit and exact analytical resolution. Setting $y_j = \beta_j^2/\hbar_B = b\kappa_j^2/\hbar_T$ fixed as $\hbar_T = 3b\hbar_B \to 0$ (`scripts/toda_edge_ba.py`, `scripts/matching_toda.py`): **(c)**

- *Head-on channel:* The exact Fermi-edge phase equals $\delta/\hbar_B$ with vanishing $O(\hbar^0)$ residual ($< 10^{-6}$ as $y_j \to \infty$), matching the small-amplitude head-on continuum limit $I_{\rm head}(\alpha_1, \alpha_2) = O(\alpha_1\alpha_2 \ln(\alpha_1\alpha_2)) \to 0$.

- *Overtaking channel ($y_1 = Y > y_2 = r^2 Y$, $r \in (0,1)$):* The Phase 2c Gamma candidate matches the Fermi-edge phase to $\lesssim 10^{-3}$ for $\eta \gtrsim 2.5$ but leaves a genuine two-body discrepancy of up to ~ 0.1 near threshold (`toda_edge.pdf`). In the semiclassical limit $Y \to \infty$ after $\hbar_T \to 0$, the Toda residual $R_0(Y, r)$ approaches a smooth non-zero $O(\hbar^0)$ function $c_0(r)$ (`matching.pdf`, `matching_tables.tex`).

- *Exact analytical resolution of $R_0(Y, r)$ (`charge_frame_master.pdf`):* Measuring dressed momentum relative to the quantum Fermi edge $P_{\rm dr}(p_j) - P_{\rm dr}(q) = P_n(\kappa_j)/a$ absorbs the $\ln \hbar_T$ ($\ln \lambda$) infrared divergence into q, shifting $u_j := \lim_{\hbar_T \to 0}(p_j - q)/\hbar_T = y_j - \frac{1}{2\pi}\ln y_j + c_{\rm edge} + O(y_j^{-1})$. In the classical edge Bethe phase $\Phi_{\rm cl}(u_1, u_2) = 2(u_1 - u_2)\ln\frac{1-r}{1+r} + 4\sqrt{u_1 u_2}$, the topological frame term $+4\sqrt{u_1 u_2} = -M_1\Pi_2/\hbar_T$ cancels the $-2/r$ Darboux singularity in $\partial\Phi_{\rm cl}/\partial u_2 = -2\ln\frac{1-r}{1+r}$ identically. Furthermore, because $v_{\rm good}(\alpha) = \cos\alpha$ decreases with amplitude while $v_{\rm bad}(\beta) = \cosh\beta$ increases with amplitude, overtaking collisions ($v_1 > v_2$) reverse the amplitude ordering ($\alpha_1 < \alpha_2$ vs $\beta_1 > \beta_2$), flipping $H_{\rm bad}(r) = -H_{\rm good}(r) = +\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$. Combining the quasi-momentum shift $d\Phi_{\rm cl}(Y, r)$ and the intrinsic loop shift $d\Phi_{\rm loop}(Y, r)$, both the $\pm 4r\delta u_1 \sim \mp\frac{2r}{\pi}\ln Y$ topological terms and the $\pm\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ logarithmic terms cancel identically, yielding the exact $Y \to \infty$ limit **(c)**

$$\boxed{\lim_{Y\to\infty} R_0(Y, r) = -\frac{4}{\pi}\big(\chi_2(r) - r\big) = -\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{r^{2n+1}}{(2n+1)^2}} \tag{3}$$

(where $+\frac{4r}{\pi}$ is the Fermi-edge reference subtraction at $r \to 0$), which matches the numerical Phase 2f intercepts ($r = 0.4, 0.5, 0.6$) to 1.2–7.0 $\times 10^{-5}$ and the r-slope to 2.9×10^{-4} (`scripts/charge_and_frame_audit.py`).

6 Route E: integrable defects and the Universal KP-Coboundary Master Function

Corrigan and Zambon (CZ) constructed integrable type-I Bäcklund defects for the Boussinesq equation **(a)** [5]. Each defect carries a single KP root $q_D = -s/(2\sqrt{3})$ ($s = \eta\sigma$, $\eta = \pm 1$), whereas a soliton carries a KP chord (p_a, p_b) connecting two roots of the KP cubic $4q^3 - q - \ell/(3\sqrt{3}) = 0$.

Semiclassical soliton–defect transmission (Phase 2c, `defect_semiclassics.pdf`). (c)

- The classical soliton–defect transmission phase is $\delta_D = \frac{2}{9}(\ell - \ell_D)\ln|z_t| - \frac{2}{3}(\Pi M_D - \Pi_D M)$ with $z_t = ((p_b - q_D)/(p_a - q_D))^{\epsilon\eta}$ and $\ell_D = (3s - s^3)/2$. The static defect adds no ordering-dependent cross terms; two-soliton + defect scattering is symplectic and factorises.

- A defect vacuum q_D is linearly stable if and only if $s = \eta\sigma < -1$ ($q_D > 1/(2\sqrt{3})$), which selects the unique ground state in each KP triple $\{q_1, q_2, q_3\}$.

- Along any edge $q_i \to q_{i+1}$ of a KP triangle where a soliton S_i on $\{q_i, q_{i+1}\}$ is absorbed by the defect, the tree-level phases satisfy exact defect fusion $\delta_D(q_{i+1})(3) - \delta_D(q_i)(3) = \rho_{{\rm tree},i}[\delta(3, S_i) + M_{S_i}\Pi_3]$.

One-loop soliton–defect phase and KP-triangle closure (Phase 2g, `oneloop_defect_phase.pdf`). (c)

- Phonon scattering on N-soliton + CZ defect backgrounds is reflectionless ($R(q) = 0$) with factorised transmission amplitude $T_{\rm tot}(q) = \big(\prod_{j=1}^{N} A(j, \mathrm{ph}(q))\big) z_{\rm ph}(q; q_D, \eta)$.

- Setting $q_D = -(\cos\gamma)/\sqrt{3}$, the defect phase-shift derivative is a *2-root* $(-\eta, +\eta)$ *dipole* $\theta'_D(q; s, \eta) = \mathrm{Re}\sum_{s'=1}^{2} c_{D,s'}\Theta'(q; \psi_{s'})$ at $\psi_{1,2} = \pm 2\pi/3 - \gamma$, whose difference across (p_a, p_b) reproduces the 3-root $(1, 1, -2)$ soliton kernel $\theta'_S(q; \alpha, \epsilon)$.

- The 4×4 defect-vacuum localised-mode determinant factors into two quadratics with closed-form eigenvalues $\omega_{1,2}(\sigma, \eta) = \frac{i}{8}[\eta\sigma\sqrt{4-\sigma^2} \pm \sqrt{3}(2-\sigma^2)]$, explaining the Levinson jumps at $s = -1$ and $s = 2$.

- Because the static defect has no translational zero mode ($\theta_D^{\rm phys}(0^{\pm}) = 0$), $I_{SD} = J_{SD}^{\rm phys}$ evaluates in closed form to a 3×2 Legendre χ_2 sum. One-loop defect vacuum shifts $(\Delta E_D^{\rm ren}, \Delta P_D)$ balance $(\Delta E_1^{\rm ren}, \Delta P_{\rm mode})$ in soliton absorption, and one-loop defect fusion $I_{SD}(3; q_{i+1}, \eta) - I_{SD}(3; q_i, \eta) = \rho_i I_{SS}(3, S_i)$ holds on every KP edge and closes around every KP triangle ($\sum_{i=1}^{3} \rho_i I_{SS}(3, S_i) = 0$), whereas the $+\frac{\pi}{2}\delta_{\epsilon_3, \epsilon_{S_i}}$ constant of J_{SS} fails to close by $\pm\pi/2$.

Universal KP-Coboundary Master Function (Phase 2h, `charge_frame_master.pdf`). (c) For *any* odd ratio kernel $K(z) = -K(-z)$, the 2×2 dipole–dipole **two-vertex master function** on the KP root space (`scripts/kp_master_and_toda_edge.py`),

$$\mathcal{F}_K(p, q) := \sum_{s_1=1}^{2}\sum_{s_2=1}^{2} (-1)^{s_1+s_2}\,\mathrm{Re}\,K\left(\tan\frac{\psi_{s_1}(p)}{2}\cot\frac{\psi_{s_2}(q)}{2}\right), \tag{4}$$

generates the soliton–defect functional as its first difference $\mathcal{I}_{SD}[K](\alpha, \epsilon; q_D, \eta) = \eta(\mathcal{F}_K(p_b, q_D) - \mathcal{F}_K(p_a, q_D))$ and the two-soliton functional as its 4-vertex second difference

$$\boxed{\mathcal{I}_{SS}[K](1,2) = \mathcal{F}_K(p_{1,b}, p_{2,b}) - \mathcal{F}_K(p_{1,a}, p_{2,b}) - \mathcal{F}_K(p_{1,b}, p_{2,a}) + \mathcal{F}_K(p_{1,a}, p_{2,a})} \tag{5}$$

guaranteeing defect fusion and KP-triangle closure for any odd $K(z)$. Splitting the one-loop kernel into $K_L = K_{\chi_2} + K_{\ln}$ with $K_{\chi_2}(z) = -\frac{1}{\pi}\chi_2(z)$ and $K_{\ln}(z) = -\frac{1}{2\pi}\ln|z|\ln\left|\frac{1-z}{1+z}\right|\mathrm{sgn}(\mathrm{Re}\,z)$ shows that at finite amplitude $\mathcal{I}_{SS}[K_{\ln}]$ is the logarithmic quasi-momentum coboundary (which cancels $H_{\rm bad}(r)$ in the small-amplitude overtaking corner), leaving the pure Legendre χ_2 coboundary $\mathcal{I}_{SS}[K_{\chi_2}] \to -\frac{4}{\pi}\chi_2(r)$.

7 Consistency checks across candidates and orders

Condition	Result	Status
Unitarity	$\delta(1,2) = -\delta(2,1)$ and $I_{SS}(1,2) = -I_{SS}(2,1)$, exact	**(c)** passed
Crossing (substitute)	$\delta_{\rm ov}(\beta_1, \beta_2 + i\pi) = -\delta_{\rm ho}(\beta_1, \beta_2)$; $I_{SD}(\alpha, -1; s, \eta) = I_{SD}(\alpha, +1; -s, \eta)$	**(c)** passed
Yang–Baxter / KP-triangle fusion	Diagonal YBE trivial; tree-level δ and one-loop I_{SS} close around all 12 KP defect triangles ($\sum_i \rho_i I_{SS} = 0$); Gamma candidate's $\pm\pi/2$ constant fails closure by $\pm\pi/2$	**(c)** decisive
Semiclassical limit vs. time delays	$\Delta x'_j = -\partial\delta/\partial P_j$ by symplectic consistency (Prop. 2.2); canonical one-loop momentum $\Delta P_{\rm can}$ obeys $d\Delta E_{\rm can} = v\, d\Delta P_{\rm can}$ with zero $\alpha \ln \alpha$ term	**(c)** passed
Bound states / bootstrap	Breather existence boundary and Bogdan–Charkina action reproduced (Prop. 2.6); one-loop string condition unshifted at fixed rapidity in Scheme A (`oneloop.pdf`)	**(c)** passed
Good equation (cusp & resonances)	One-loop phase I_{SS} finite and smooth across all 3 resonance lines ($A_{12} \to 0, \infty$); jumps by $\pm\pi$ at the cusp $\alpha = \pi/3$ where a heavy-branch instability appears (`oneloop_phase.pdf`)	**(c)** passed
Quantum Toda Fermi-edge limit	Tree-level $\delta_T \to 3b\,\delta_{\rm bad}$; $O(\hbar^0)$ head-on residual $= 0$; $O(\hbar^0)$ overtaking residual $\lim_{Y\to\infty} R_0(Y, r) = -\frac{4}{\pi}(\chi_2(r) - r)$ matches Phase 2f table to $< 7.1 \times 10^{-5}$ (`charge_frame_master.pdf`)	**(c)** passed

8 Conclusions and ranked open problems

Conclusions (Phases 2a–2h).

1. *Tree-level ($O(\hbar^{-1})$) and one-loop ($O(\hbar^0)$) two-soliton scattering are completely solved and reconciled.* At $O(\hbar^{-1})$, the phase is (1), where the ordering-dependent topological cross term $d\Pi_i \wedge dM_j$ is essential. At $O(\hbar^0)$, the continuum phase is the 4-vertex KP coboundary (5) of the dilogarithm kernel $K_L = K_{\chi_2} + K_{\ln}$, whose small-amplitude Fermi-edge limit reproduces the quantum Toda Bethe ansatz residual $-\frac{4}{\pi}(\chi_2(r) - r)$ to $O(10^{-5})$. **(c)**

2. *Both simple candidates (the Gamma ansatz and the pure exponential) are ruled out at $O(\hbar^0)$.* A single Gamma factor $\Gamma(1 + i\eta)/\Gamma(1 - i\eta)$ in $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ has only rational/Bernoulli asymptotic corrections in η and cannot produce the dilogarithm $\chi_2(z)$ at $O(\hbar^0)$; moreover, its $\pm\pi/2$ constant violates KP-triangle defect fusion. **(c)**

3. *Every physical soliton observable is a KP-chord difference.* Both at $O(\hbar^{-1})$ ($\ln A_{12}$ as the 4-vertex cross-ratio of $\ln|p - q|$) and at $O(\hbar^0)$ ($\mathcal{I}_{SS}[K]$ as the 4-vertex second difference (5) of the 2×2 dipole–dipole master function $\mathcal{F}_K(p, q)$), two-soliton scattering is governed by a two-vertex master function on the KP root space. **(c)**

Updated ranked next steps (after Phase 2h).

1. **Analytical $O(1/Y)$ (two-loop, $O(\hbar^1)$) expansion of the quantum Toda Fermi-edge Bethe ansatz.** Carry the analytical Fermi-edge expansion of `scripts/toda_edge_ba.py` one order further in $1/Y$ (combining the $O(\hbar/D)$ Stirling/Bernoulli tail of the Toda loggamma kernel, the $O(1/y_j)$ quasi-momentum shift, and the rank-1 Hessian of $\Phi_{\rm cl}$) to derive a closed-form formula for the two-loop coefficient $c_1(r)$ extracted numerically in `scripts/kp_master_and_toda_edge.py` ($c_1(0.4) = -0.00708$, $c_1(0.5) = -0.02099$, $c_1(0.6) = -0.04822$).

2. **All-loop KP master kernel $K(z;\hbar)$ and quantum dilogarithm test.** Use the exact tree-level (ln), one-loop (χ_2), and two-loop ($c_1(r)/Y$) data together with the Universal KP-Coboundary representation (5) to test whether the all-loop master function $\mathcal{F}_K(p,q;\hbar)$ is given by a quantum dilogarithm (Faddeev–Volkov / Barnes double Gamma function) on the KP root space.

3. **Ring quantisation with ϕ-winding and twisted boundary conditions.** Formulate the periodic N-soliton Bethe equations incorporating the topological cross term $-\frac{2}{3}(\Pi_i M_j - \Pi_j M_i)$ and the one-loop phase $\mathcal{I}_{SS}[K_L]$.

4. **Non-ultralocal quantum inverse scattering** (Maillet) for the Boussinesq Lax pair (lowest priority).

A Phase 2 Write-ups and Verification Scripts

All write-ups are in `phase2/*.pdf` and all scripts are in `phase2/scripts/*.py` (plus `corrigan-zambon/defect_` each script ends with `ALL CHECKS PASSED`.

- **Phase 2a (`report.pdf`):** `scripts/symplectic_scattering.py`, `scripts/semiclassical_phase.py`, `scripts/fit_R.py`, `scripts/consistency_checks.py`, `scripts/routes_BC_checks.py`.
- **Phase 2b (`toda_route.pdf`):** `scripts/toda_bethe_classical.py`, `scripts/toda_bethe_quantum.py`, `scripts/toda_ba_vs_label.py`, `scripts/toda_scaling_fast.py`, `scripts/toda_quantum_correction.`
- **Phase 2c (`smatrix_ansatz.pdf`, `defect_semiclassics.pdf`):** `scripts/smatrix_ansatz.py`, `scripts/smatrix_boot.py`, `scripts/defect_smatrix_good_continuation.py`, `corrigan-zambon/defe`
- **Phase 2d (`toda_edge.pdf`, `oneloop.pdf`):** `scripts/toda_edge_ba.py`, `scripts/toda_edge_twoterm.p` `scripts/oneloop_step0_wellposedness.py`, `scripts/oneloop_phase_shifts.py`, `scripts/oneloop_s` `scripts/oneloop_breather.py`.
- **Phase 2e (`oneloop_phase.pdf`):** `scripts/oneloop_good_phonon_scattering.py`, `scripts/oneloop_g` `scripts/oneloop_good_limits_cusp_resonance.py`.
- **Phase 2f (`matching.pdf`):** `scripts/matching_conventions.py`, `scripts/matching_toda.py`, `scripts/matching_analysis.py`.
- **Phase 2g (`oneloop_defect_phase.pdf`):** `scripts/oneloop_defect_scattering_and_stability.py`, `scripts/oneloop_defect_phase_and_fusion.py`.
- **Phase 2h (`charge_frame_master.pdf`):** `scripts/charge_and_frame_audit.py`, `scripts/kp_master_an`

Failures along the way (Phase 2a; later phases record their own failures in each write-up).

- The direct symbolic simplification of the symplectic identity timed out, so we switched to 40-digit numerics.
- The good-equation phase first used the wrong sign of ℓ, and the test failed until it was corrected.
- A tolerance set below double precision caused spurious failures.
- The KdV-limit comparison with Bonnemain–Doyon *does not* agree. This is recorded as a finding, not a passed test.

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